Gravity above the surface

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use distance from a planet's center to compare fields at altitude.

SUCCESS CRITERIA - I can...

1. Use r=R+h rather than altitude alone.
2. Compare fields with the inverse-square relation.
3. Explain why orbit is not a zero-gravity region.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "altitude" and "radius" to explain your answer. Sentence starters are optional.

altitude: height above a reference surface

radius: distance from the center to the edge of a circle or sphere

field: a description of an influence at each location

01 Warm-up: bring back an earlier idea

For fixed masses, what separation multiplier makes gravity **1/9** of its original value? What multiplier is needed if both masses also become **5** times as large?

Word help: separation = distance between the specified objects or centers

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: ① ② ③

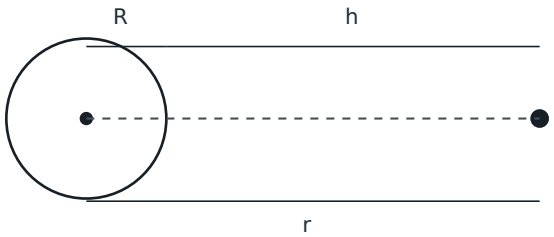
02 Worked example: explain the prediction

Reference: $r = R + h$; $g_h/g_{surface} = [R/(R + h)]^2$.
R is planet radius; h is altitude. g_h is field strength at altitude h.

Worked reasoning: At altitude h=**2** R above a planet's surface, center distance is r=R+h=**3** R. A surface field of **18 N/kg** becomes **18/9=2 N/kg**.

Your turn: Draw the radius and altitude as different line segments. **Mark the full center-to-object distance**.

Sentence starter: Both ... . They differ in ... because ... .



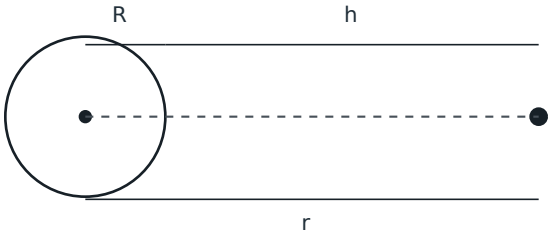
Measure r from the center: r = R + h.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A satellite is one planet-radius above the surface.
1. **Label R**, h and r on a sketch.
2. What is r in units of R?

Sentence starter: My design includes ... . It does what is needed by ... .



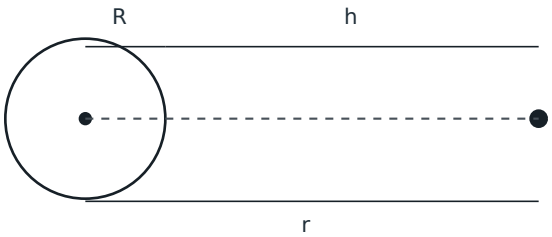
Label surface radius R, altitude h and center distance r.

Marzano check - circle: ① ② ③

04 Apply the physics

For Earth use$R=\mathbf{6.4\times10^6\ m}$, $g_{surface}=\mathbf{9.8\ N/kg}$. **Find g** at $h=\mathbf{4.0\times10^5\ m}$ and at h=R.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Explain why an astronaut at modest orbital altitude can float even though gravity is only slightly weaker there.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **4 kg** box has μ_s=**0.50** and μ_k=**0.40** on a level floor; g=**9.8**. **Compare pushes of 8 N** and **24 N**, starting from rest. **Find actual friction** and acceleration.
Draw the forces on the chosen object and **label their sources**.

Word help: friction = a contact force opposing relative sliding or its tendency

Sentence starter: Both … . They differ in … because … .

Add and label force arrows.

Marzano check - circle: 1 2 3

07 Exit check

At what altitude in units ofR is field one quarter of surface value? At what altitude is it one ninth?

Sentence starter: I notice … . This supports … because … .

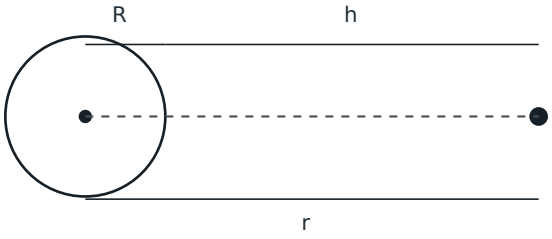
Field / surface field	r / R	h / R
___	___	___
___	___	___

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

1. **Find the altitude h**/R where a spherical planet's field is **10%** of the surface value.
2. **Explain why h**/R is not √**10**.

Sentence starter: My design includes … . It does what is needed by … .



Marzano check - circle: 1 2 3